1

**Interviewer**

2 Could you please briefly introduce yourself and provide a bit of background on your experience working with genomic data?

3

13 **Participant**

4 Hi. I'm a first year PhD student who is starting to work with genomic data.
14 Some of the background that I've had is using the AllofUs research platform and their genomic data. They have synthetic tables that you can look at. They also have different phenotypes you can look at that are synthetically made.

5:3 I'... Specific software t...

5

**Interviewer**

6 What do you remember about your first experiences working with that data and the tutorials? How did you get started with it all?

7

**Participant**

8 After I was involved with the project, they sent me an email saying that they had a new user orientation. I completed the orientation and from that I was able to find out that they had data sets available to use or to get more familiar with everything. So, like, before that, I wasn't familiar at all with how they set up AllofUs research at all with the data.

5:1 After I wa... Resources used w...

9

**Interviewer**

10 And what were these tutorials like? How are they structured?

11

**Participant**

They're very simplistic. It was mostly like a PowerPoint with a person talking over it. It was a video that's, like two minutes long, describing each or each video was two minutes long, describing each thing, such as how to go from a workspace to the cohort to the dataset and how to get through each one, which was I actually kind of liked how simple it was instead of making it overly complex.

5:2 And what were these tutorials like? ... Resources used w...

**Interviewer**

Could you give a brief description on what the workflow is like for AllofUs?

15 **Participant**

16 It's all online, but I think all the data is in the cloud. So you can take what you need from the cloud into your workspace that they have online for you to look at, and it's on your computer as well.

17

**Interviewer**

18 Got you. Other Than AllOfUs, are there any other analysis tools that you're starting to learn or you've utilized in the past?

19

27 **Participant**

20 I'm starting to use Cypress, which is like an HPC sort of thing, and so I've
28 been trying to use that a little bit. It's also kind of like a cloud thing because it's like on the Linux Unix type thing where you have to have it terminal, and then you go through a specific shell to get to the information that you need.

5:4 I'm starting to ... | Specific software t...

21

**Interviewer**

22 How did you feel when you were beginning to work with the AllOfUs platform after you completed the tutorial?

23

**Participant**

After the tutorial I felt a little bit better about it. I did feel a little bit overwhelmed with just looking at everything beforehand, but afterwards I felt like more confident that I could use their workspace more.

5:5 After ... | Cognitive reaction | Emotional reaction

**Interviewer**

Did you have any concerns or questions when you were utilizing either All Of US or Cyprus? Was any information missing?

**Participant**

AllOfUs didn't really have anything that I thought that they were missing. I thought they were pretty clear on how to use everything and they're very informative. And then they also add a more detailed help support system in terms of like, if you did get lost in trying to understand how to use one of their specific tasks that they had. For Cypress, I would say I did have some concerns for it because they also had like a workshop series for it at Tulane. But I think if I didn't have that, like, workshop, I would have been

5:6 AllOfUs didn't ... | Benefits of specifi... | Cognitive reaction

5:7 For C... | Barriers of specific... | Cognitive reaction

28 clueless. Yeah, because it wasn't made clear on their Wiki page because they have an entire Wiki page of how to use Cypress, but it's a little bit harder to navigate than AllOfUs, I would say.

5:7

29

**Interviewer**

30 What do you see as positives about utilizing All Of US and Cypress?

31

**Participant**

I think the positives for All Of US is that it's very I guess it's like strict on everything since it is online. It's like on a website, there is that like fear that someone could possibly be able to look at it at all these different data sets just because it's like by hacking or something. But they do have a lot of positives of the double authentication and different ways of making sure that these data sets are used for a purpose of some sort, so you can't just
33 log in and access them like one specific person data set. So I like how it is. Very prided, I guess, on all of us. And then a positive for Cypress is that it can hold a lot of information and you can access it basically on any
34 computer that you want as long as you have like, a terminal of some sort. And then you know your login information. So it's also secured too, because that's not on a website, but it's based up to figure out where you need to go in the terminal to get there. So I guess I just like the security,
35 the positive side of that.

5:8 I ...
Benefits of specifi...
5:9 It's lik...
Barriers of specific...
5:10 they do have...
Benefits of specifi...
5:11 then a positive for Cy...
Benefits of specifi...

**Interviewer**

36

Yeah, it's always nice to make sure that the data is not easily tampered with. What about some negatives that you see from both platforms?

**Participant**

Some negative, maybe for Cypress, I guess I'm a little bit hesitant on using it just because I know there's certain Linux commands that once you do, like, hit Enter and have that command run, there's no undo button. So you can't really go back unless you have a copy of some sort. Like, let's say you remove a directory or remove a file of some kind. You can't really get it back unless you have a copy or backup of some kind. I'm not too sure about AllofUs just yet. I haven't seen like too many negatives about it, but I
37 haven't worked with it too much.

5:12 for Cypress, I gue...
Barriers of specific...

**Interviewer**

38 Got you. Has utilizing AllOfUs or Cypress influenced your cost of your experiments in any way?

39

**Participant**

40 I know AllOfUs gives you a set, like they give you a credit line of some sort when you start using it. I don't know how cost efficient it is, but I feel like it
49 should be since all the data is just in one place.

5:13 I kn...

Impact on cost/up...

41

**Interviewer**

50

42 Other than the positives and negatives, has utilizing AllOfUs or Cypress influenced any other aspect of your experience of being a researcher?

43

**Participant**

44 I did look up some things online because AllOfUs uses Python or R, you have the option of choosing either language, but sadly, I don't know those languages yet. So I did have to look up some little syntax, little cheat sheets for R because I thought that would be helpful. I do want to take a course on R just because I haven't understood it yet. And then for Cypress, I've already taken a Linux course at Florida State, so I'm more familiar with how to use that. But if I didn't have that information beforehand, I think I would have been totally clueless on how to use Cypress.

5:14 I did look up some thi...

Overall impact

45

**Interviewer**

46 Do you have any recommendations for changes or additions that would have aided you in your initial on boarding or your future research?

47

**Participant**

Probably for Cypress, at least maybe make it more clear on the Wiki page that they have, but that's just like a preference for me. Their Wikipedia page was not great to manage because there's a lot of like, hyperlinks, but I don't think you would necessarily know which one to choose if you were just starting out.

5:15 Probably for ...

Suggested improv...

**Interviewer**

Okay, and a final question, more of like a theoretical question. As the NHGRI and other genomics research organizations move more towards cloud computing, do you anticipate any challenges or concerns with doing research in the cloud?

51 **Participant**

52 I guess one concern for the cloud is I guess who necessarily has access to the cloud because it's like if we only are allowed to get like, certain portions of it. But is there someone who's maintaining trying to keep the cloud secure? It's like, is there a way for someone to access the entire cloud? Yeah, if there's a concern or if there's a possibility of someone accessing the entire data cloud, I mean, it would be way too much to even look at, but that could be a concern.

5:16 I guess one concern f...

Challenges/conce...

53

**Interviewer**

54 That's an important concern. Cool. I think that those are all the questions that I had. I'm going to stop the recording now.